

WASP-10b

R-1864

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_160923 · created 2026-08-03T09:53:30+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457651.6171 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.062
Transit depth (Rp/Rs)^2	2.34 +/- 1.89 [%]
Orbital Inclination (inc)	85.65 +/- 1.87
Airmass coefficient 1 (a1)	1.456 +/- 0.066
Airmass coefficient 2 (a2)	-0.24 +/- 0.21
Scatter in the residuals of the lightcurve fit is	4.6 %
Best Comparison Star	#1 - [593, 165]
Optimal Aperture	10.8
Optimal Annulus	11.72
Transit Duration (day)	0.0 +/- 0.0

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.153 ± 0.062 vs 0.1592 (deviation 0.1σ)
Duration fit vs published	0.0 d vs 0.0946 d (deviation infσ)
Mid-time O–C	-4448.2 min
Depth SNR	2.5
Residual scatter	4.6 %

Light curve

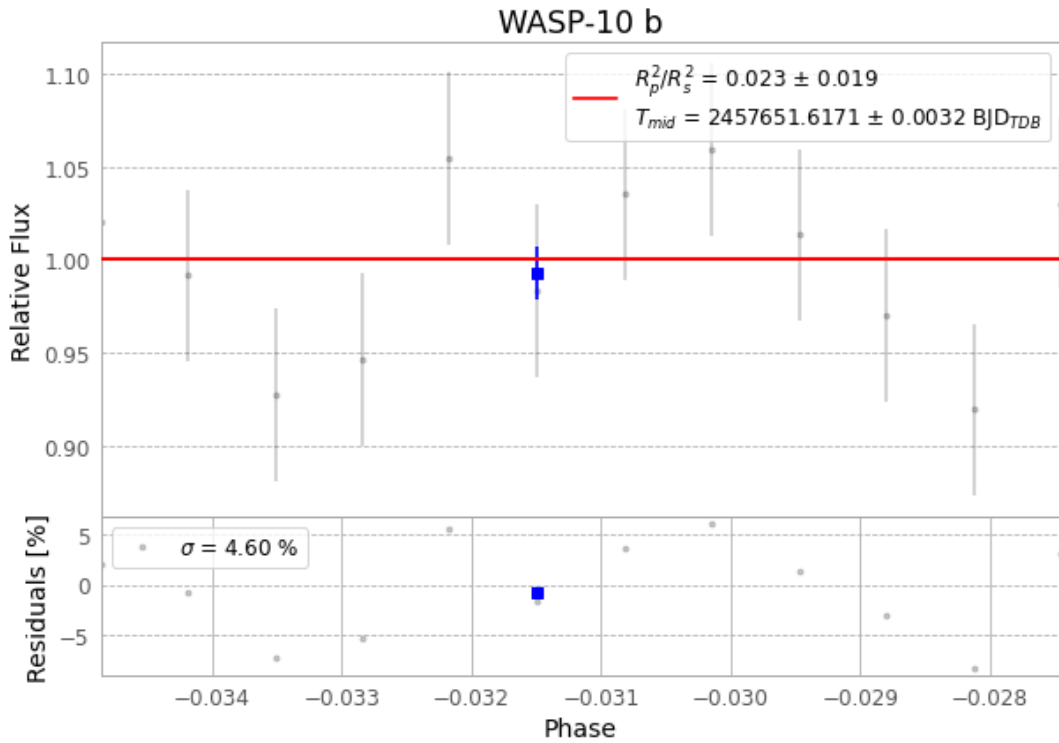


Figure 1 — FinalLightCurve_WASP-10 b_2016-09-22.png (SHA-256 37668268368b05bf...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [501, 227], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e8a58b109f72fb74...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

115 FITS frames (76 MB), WASP-10160923021813.FITS → WASP-10160923080012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-160923.log (d9e0e80d6c2f...), FinalLightCurve_WASP-10 b_2016-09-22.png (37668268368b...), FinalLightCurve_WASP-10 b_2016-09-22.csv (3d8181aa2f1a...)

Compute resources

Wall time 300.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 09:53Z.